

## Assignment 8: Equations

### DO NOT WRITE ON THIS PAPER

#### Part A

The following is an expression:  $(0.5x-5)$ . How do you know it is an expression?

Evaluate it at  $x=0$  and  $x=10$ . Show your steps.

#### Part B

The following is an equation:  $0.5x - 5 = 0$ . How do you know it is an equation?

Solve the equation for  $x$ . Show your work.

#### Part C

Consider the equation:  $ax - b = cx - d$ .

Fill in the values for **a**, **b**, **c**, and **d** by rolling dice. Your first roll is **a**, your second roll is **b**, and so on. Solve the equation for  $x$  and explain your steps.

*Example: if you roll a 6, 4, 2, and 5, your equation would look like  $6x - 4 = 2x - 5$*

### Question 1

**Big Question:** What are you allowed to do to an equation?

Here is an equation:  $0.2x - \frac{1}{13} = 12x - 3$ . Below are a few examples of me "doing something" to an equation. Next to each, write **allowed** or **not allowed** and explain your reasoning.

a.  $123(0.2x - \frac{1}{13}) = 123(12x - 3)$

b.  $(0.2x - \frac{1}{13})^7 = (12x - 3)^6$

c.  $(0.2x - \frac{1}{13}) + 4 = 123(12x - 3) - 4$

### Question 2 Work

Solve for x. Include an explanation of each step.

$$\frac{x}{5} + 5.3 = -7.2$$

### Question 2 Answer

Write your answer  
in your Class Notes

### Question 3 Work

Solve for y. Include an explanation of each step.

$$-6 = \frac{5y-17}{-3}$$

### Question 3 Answer

Write your answer  
in your Class Notes